Recommendations for 180nm AND / XOR Gate Additions

Sanskar Jain | Roll No. 2023486 | Scope: AND and XOR gates only

Work Profile

Scope
15 variants

Depth
Focused low-drive gap analysis

Recommendation
Drive-strength addition

Evidence
Normalized delay comparison

Analysis Basis

- Focused on 180nm vs 65nm normalized delay comparison in the low-drive region for the AND and XOR families.
- Studied the gap between X1 and X2 to determine whether the current ladder is too coarse for marginally critical paths.
- Kept the analysis centered on drive-granularity need rather than on missing logic-function variants.

Key Recommendations

1) Add X1.5 (Primary Recommendation)
Creates a true intermediate remap target between X1 and X2, reducing oversizing in the low-drive region.

2) No new functionality addition proposed
The observed weakness is a drive-strength gap, so library effort should remain focused on X1.5 characterization first.

Reasoning / Evidence

- The 180nm normalized delay points stay above the 65nm trend in the low-drive region, indicating insufficient granularity.
- X1.5 is therefore the most defensible intermediate step for both AND and XOR families.
- This workstream is intentionally narrow and focused, with evidence coming from the normalized low-drive comparison itself.

Validation Note: These are intermediate analysis-phase recommendations, not sign-off claims. Final acceptance should come from the next design-validation phase, including simulation and/or hardening, and from measured remapping impact on timing, leakage, area, capacitance, and implementation overhead.